Repo Download:

Verify Installed:

- `which sshpass git python3 ssh`
- `which sshpass` /usr/bin/sshpass
- `which git` /usr/bin/git
- `which python3` /usr/bin/python3
- `which ssh` /usr/bin/ssh

If not installed:

- Rocky
  - `sudo dnf install -y sshpass`
  - `sudo dnf install -y git`
  - `sudo dnf install -y python3`
  - `sudo dnf install -y openssh-clients`
- Verify
  - `ssh -V`
- Ubuntu/Debian
  - `sudo apt update`
  - `sudo apt install -y sshpass`
  - `sudo apt install -y git`
  - `sudo apt install -y python3`
  - `sudo apt install -y openssh-client`
- Verify
  - `ssh -V`

Download Repo:

- `git clone https://github.com/CCDC-Century-College/CCDC-2026.git /root/.reference-1`
- `git clone -b beta https://github.com/CCDC-Century-College/CCDC-2026.git /root/.reference-2`

Restrict Access:

- `sudo chown -R root:root /root/.reference-<#>`
- `sudo chmod -R 700 /root/.reference-<#>`

Run Scripts:

- `cd /root/.reference-<#> && ls -la`
- `python3 linux_internal_credential_rotation.py`

Rocky Linux x9.x/Ubuntu x22.x/Debian x12.x Hardening

Step#1: PIDs

- A: Username Check
  - `getent passwd <username>`
  - `id <username>`
  - `who -u`
  - `ps -u <username> -f`
- B: Red Flags (Notes:1B)
  - `readlink -f /proc/<PID>/exe`
  - `kill -TERM <PID>`
  - `passwd -l <username>`
  - `sleep 3`
  - `ps -p <PID> -f`
  - `kill -9 <PID>`

Step#2: Network Ports

Rocky

- A: Listening Ports (Red Flags Notes:2D)
  - `ss -tulpn` (PIDs:1B)

Ubuntu/Debian

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Step#3: Persistence

- A: Cron Jobs/Scheduled Tasks
  - `crontab -l`
  - `sudo crontab -u root -l`
  - `ls -la /etc/cron* /var/spool/cron/`
- B: Red Flags
  - `cat /etc/cron.<?>/<filename>` (Red Flags Notes:3A) (YES:3B) (NO:4)
  - `pgrep -fl <script_name>` (PIDs:1B)

Step#4: Persistence systemd

- `systemctl list-unit-files --state=enabled`

- A: States Investigation
  - `systemctl cat <servicename>` (Red Flags Notes:1B) (YES:4A) (NO:4B)
  - `systemctl stop <servicename>`
  - `systemctl disable <servicename>`
  - `pgrep -fl <binarynamefromexecstart>` (PIDs:1B)
- B: Activates Column Investigation
  - `systemctl list-timers --all` (Red Flags Notes:4B YES:B1) (NO:Next Timer)
    1. `systemctl cat <activatedservice>`
       - `systemctl stop <timername>`
       - `systemctl disable <timername>`
- Verify: Timer was disabled.
  - `systemctl is-enabled <timername>`

Step#5: Firewall/Network Sockets

- A: Firewall Status
  - `systemctl status firewalld`
  - `firewall-cmd --state`
  - `firewall-cmd --list-all` (Red Flags Notes:5A)
  - `systemctl enable --now firewalld` (If firewall is disabled)
- B: Manual IP Blocking
  - `firewall-cmd --permanent --add-rich-rule='rule family="ipv4" source address="<IPaddress>" reject'`
  - `firewall-cmd --reload`
- Verify: IPs are blocked
  - `firewall-cmd --list-rich-rules`

Ubuntu/Debian

- A: Firewall Status UFW
  - `sudo ufw status verbose`
  - `sudo ufw show added`
  - `sudo ufw enable` (If firewall disabled)
  - `sudo ufw logging medium` (If logging low)
- B: Manual IP Blocking
  - `sudo ufw status numbered`
- C: Live Log Monitor
  - `sudo tail -F /var/log/ufw.log |fold -s -w $(tput cols)`

Quick CLI-Commands:	**Function:**
`passwd <username>`	Change password
`id`	Show user UID/GID/group
`sudo –i`	Switch to root shell
`su -`	Switch to root account
`w`	Active users
`who`	Logged in users
`last -a`	Login history
`pstree -p`	View process parent/child tree
`ps -o pid,ppid,user,cmd -p <pid>`	Find parent PID and owner
`pgrep -af <process>`	Locate process and PID
`systemctl --failed`	Failed services
`systemctl --type=service --state=running`	Running services
`systemctl list-unit-files --type=service`	Installed services
`systemctl status <service>`	Service Health
`systemctl is-active <service>`	Verify Running
`journalctl -u <service>`	Service logs
`journalctl -xe`	Recent system errors
Quick CLI-Commands: Rocky	**Funtions:**
`grep '^wheel:' /etc/group`	View wheel group members
Quick CLI-Commands: Ubuntu/Debian	**Function:**
`grep '^sudo:' /etc/group`	View sudo group members

Quick Recovery CLI-Commands:

Recovery:	**Action Previously Taken:**
`passwd -u <user>`	Lock User
`systemctl restart <service>`	Kill –9 <PID>
`systemctl enable --now <service>`	Disable Service
`systemctl enable --now <timer>`	Disable Timer

Quick Recovery CLI: Rocky (firewalld)

`sudo firewall-cmd --list-rich-rules`	View Rules
`sudo firewall-cmd --permanent --remove-rich-rule='rule family="IPv4" source address="<IP_ADDRESS>" reject'`	Remove IP Block
`sudo firewall-cmd --reload`	Reload Firewall
`sudo firewall-cmd --list-rich-rules`	Verify Firewall

Quick Recover CLI: Ubuntu/Debian (UFW)

`sudo ufw status numbered`	View Numbered Rules
`sudo ufw delete <#>`	Delete Rule Number
`sudo ufw status numbered`	Verify Firewall